

Database Installation Clarification

Project: RoomRadar QC

Purpose

This document clarifies how the RoomRadar QC database installation script should be executed. It specifically explains where the command `SOURCE install_database.sql;` should be run and how it relates to the project database setup.

1. Database Setup Overview

RoomRadar QC uses a MySQL database to store the data needed by the Flask backend.

The database stores:

Table	Purpose
semesters	Stores semester names uploaded by the admin.
rooms	Stores classroom information such as room code, building, and capacity.
schedules	Stores class schedule records connected to rooms and semesters.

The React frontend does not connect directly to the database. The Flask backend handles all database communication.

The overall data flow is:

React/Vite Frontend



Flask Backend API



MySQL Database

2. Purpose of install_database.sql

The install_database.sql file contains SQL statements used to create or install the database structure required by RoomRadar QC.

This file may include SQL statements such as:

```
CREATE DATABASE IF NOT EXISTS school_project;  
USE school_project;
```

```
CREATE TABLE IF NOT EXISTS semesters (...);  
CREATE TABLE IF NOT EXISTS rooms (...);  
CREATE TABLE IF NOT EXISTS schedules (...);
```

The exact SQL statements depend on the final version of the project schema, but the purpose of the file is to prepare the database so the backend can store and retrieve semester, room, and schedule data.

3. Where to Execute **SOURCE** **install_database.sql;**

The command:

```
SOURCE install_database.sql;
```

should be executed inside a MySQL client session after connecting to the MySQL server.

The SOURCE command is a MySQL client command. It tells MySQL to read and execute the SQL statements inside the .sql file.

4. Example: Running the SQL Script Locally or

on a Server

First, open a terminal and connect to MySQL:

```
mysql -u darshan -p
```

After entering the MySQL password, the prompt should change to something like:

```
mysql>
```

Then, from inside the MySQL prompt, run:

```
SOURCE install_database.sql;
```

If the file is inside a folder called database, then run:

```
SOURCE database/install_database.sql;
```

If the file is located somewhere else, use the correct path to the file. For example:

```
SOURCE /path/to/database/install_database.sql;
```

5. Example Full Setup Flow

A typical setup flow would look like this:

```
cd database
```

```
mysql -u darshan -p
```

Then, inside the MySQL prompt:

```
SOURCE install_database.sql;
```

After the script runs successfully, the required tables should exist in the MySQL database.

6. How to Verify the Database Was Installed

After running the script, the database can be checked from the MySQL prompt.

Show available databases:

```
SHOW DATABASES;
```

Select the project database:

```
USE school_project;
```

Show the tables:

```
SHOW TABLES;
```

Expected project tables include:

semesters

rooms

schedules

To inspect a table structure:

```
DESCRIBE rooms;
```

```
DESCRIBE schedules;
```

```
DESCRIBE semesters;
```

7. Relationship Between SQL Script and Flask Backend

The SQL script prepares the database structure.

The Flask backend uses that database structure to:

- Insert uploaded semester records
- Insert room records
- Insert schedule records
- Search for rooms without class conflicts

- Retrieve schedule details for a selected room

The backend connects to MySQL using environment variables, such as:

```
DB_HOST=...  
DB_USER=...  
DB_PASSWORD=...  
DB_NAME=school_project  
DB_PORT=3306
```

8. Local Development vs Deployed Database

Local Development

During local development, we can connect to the remote MySQL database using the MySQL client and run:

```
SOURCE install_database.sql;
```

This prepares the database for local backend testing.

Deployed Version

In the deployed version, the backend connects to the MySQL database hosted on EC2. The SQL installation script should be executed on that MySQL server through a MySQL client connection.

For example, a developer or admin can connect to the EC2-hosted MySQL database from an authorized machine and then run the script from inside the MySQL prompt.